COMMITTEE ON DATA
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Advancing cross-domain data integration for global development - learnings from the WorldFAIR project

Simon Hodson, CODATA

WorldFAIR

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IASSIST Conference 2023
Philadelphia, PA, USA and Online
2 June 2023

WorldFAIR: Global cooperation on FAIR data policy and practice

- Two-year project to advance implementation of the FAIR principles in a range of disciplines, or cross-disciplinary research areas.
- Funded by the European Union, HORIZON-WIDERA-2021-ERA-0 — Project: 101058393
- Two years from 1 June 2022.
- Nineteen partners: https://worldfair-project.eu/
- Institutions in France, Belgium, Cyprus, Denmark, Germany, Ireland, Norway; Australia, Brazil, Kenya, New Zealand, USA; UK.
- Includes **authoritative international entities** (e.g. IUPAC, OneGeochemistry, GBIF, ODIS); connections with important projects or standards organisations (e.g. NanoCommons, DDI Alliance, OHDSI, TDWG, SalUrbAL).
- Important partnership between CODATA and Research Data Alliance.
- Builds on CODATA work for ISC: 'Making Data Work for Cross-Domain Grand Challenges': https://council.science/actionplan/making-data-work-for-grand-challenges/

WorldFAIR Partners

AuScope

Australian National University

AVI

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dri Digital Repository of Ireland Taisclann Dhigiteach na hÉireann

Embrapa

FAPED

GBIF

HiveTracks know your bees!

IUPAC

KALRO Kenya Agricultural & Livestock Research Organization

LONDON SCHOOL of HYGIENE &TROPICAL MEDICINE

Meise Botanic Garden

NovaMechanics Cheminformatics & Nanoinformatics Excellence

RDA RESEARCH DATA ALLIANCE

SALURBAL Salud Urbana en América Latina

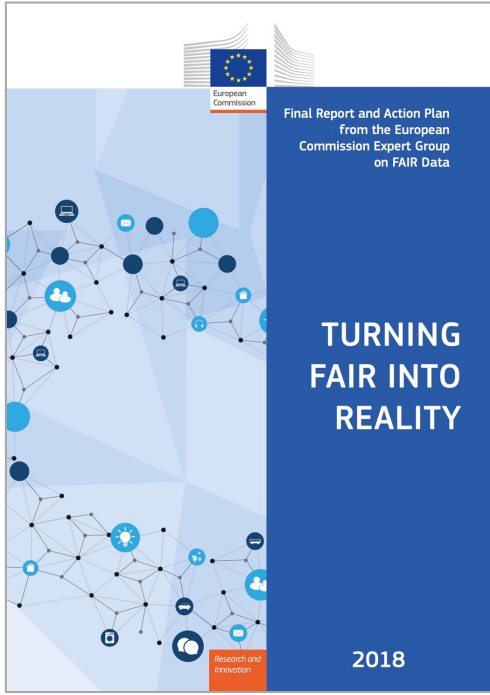
7P9 Seven past nine

Tonkin+Taylor

UNIVERSITY OF BIRMINGHAM

Interoperability Frameworks

- Among the most important, but most challenging, recommendations of the **Turning FAIR into Reality** report, is R.4:
- ‘Develop **interoperability frameworks** for FAIR sharing within disciplines and for interdisciplinary research: Research communities need to be supported to develop interoperability frameworks that define their practices for data sharing, data formats, metadata standards, tools and infrastructure. To support interdisciplinary research, these interoperability frameworks should be articulated in common ways and adopt global standards where relevant.’
- Very strong focus on the I and the R of FAIR.
- Core driver of CODATA-ISC Decadal Programme and WorldFAIR project



Making Data Work for Cross-Domain Grand Challenges

- **CODATA, established by the ISC, has a specific mission and role.** CODATA and ISC members include International Scientific Unions, and other global science organization. Strong links with the UN System.
- **ISC Action Plan entrusts CODATA with an initiative 'Making Data Work for Cross-Domain Grand Challenges': establish a global (decadal) programme to address these issues.**
- The major, pressing global scientific and human issues of the 21st century can **ONLY** be addressed through **research that works across disciplines to understand complex systems**, and which uses **interdisciplinary and transdisciplinary** approaches to turn data into knowledge and then into action.
- **Preparatory Phase from 2017:** exploring technical issues and case studies through Dagstuhl workshops, TGs and WGs, funded projects etc, to understand the challenges and prepare the programme. Developed some of the WorldFAIR methodology and approach.
    - **Cross-Domain Interoperability Framework:** units, vocabularies, data structure, data description, provenance…
    - **Case Studies:** in a range of domain and cross-domain research areas: included infectious diseases/epidemiology; urban health; DRR; Task Groups in Agriculture, Nanomaterials, etc.
- **WorldFAIR Project:** takes this approach forward in the context of an EU-funded project.

Preparatory Activities

1. **Series of exploratory workshops**
   - Two workshops in 2017, in Paris (ISC-CODATA) and London (Royal Society) launched the activity.
   - Series of Dagstuhl Workshops: 2018, 2019, 2021 (cross-domain and vocabularies) and 2022!
2. **Related outputs and activities**
   - DDI Alliance work on DDI-CDI
   - Cox et al. group (formed at Dagstuhl) produced '10 Simple Rule for Making a Vocabulary FAIR' https://arxiv.org/abs/2012.02325
   - **IUSSP-CODATA WG on FAIR Vocabularies in Population Science:** report https://doi.org/10.5281/zenodo.7818157; and webinar https://codata.org/fair-vocabularies-in-population-research/
   - CODATA TG on DRUM (Digital Representation of Units of Measure): https://codata.org/initiatives/task-groups/drum/
3. **Case Study approach**
   - Including infectious disease/epidemiology work on Ebola, COVID; urban health (including ISC UHWB Programme and Salurbal); DRR including ISC activities and CODATA TG; case studies from DDI-CDI report; and more.

The Role of DDI-CDI in EOSC: Possible Uses and Applications

The Role of DDI-CDI in EOSC: Possible Uses and Applications

Authors

Arofan Gregory, Simon Hodson, and Joachim Wackerow

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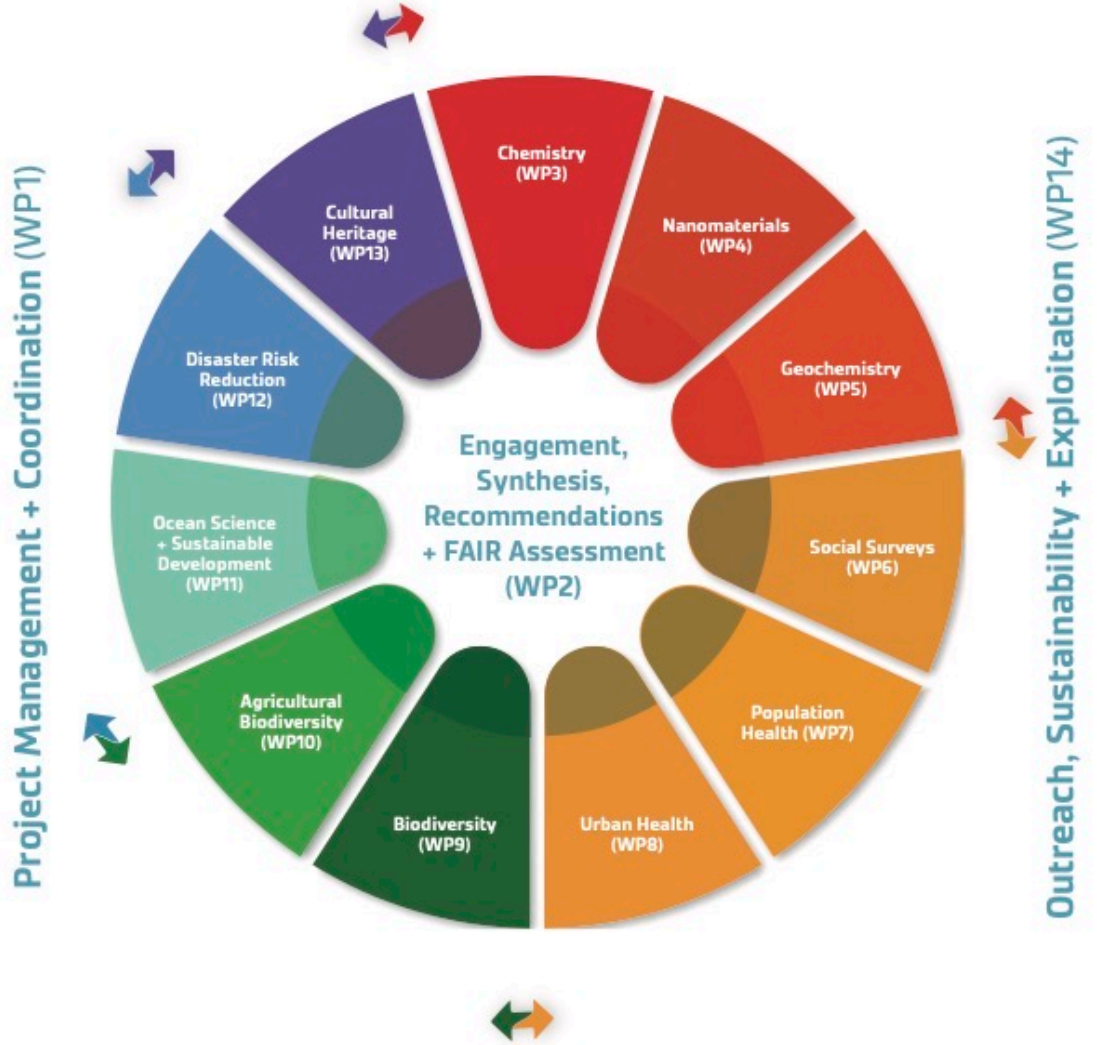
- DDI-CDI (Cross Domain Integration) is designed to interface with other standards and to help interoperability and integration of data between different data types, standards, formats.
- EOSC co-creation project to explore uses and applications and make recommendations for EOSC and the specification.
- **The Role of DDI-CDI in EOSC: Possible Uses and Applications, final report https://doi.org/10.5281/zenodo.4707263**
- Report examines the challenge for EOSC: issues of scale and the challenge of cross-domain data reuse.
- Presents examples of how DDI-CDI can be used in data integration and cross domain use cases; describes how DDI-CDI can interact with other standards.
- Series of webinars to assist review of specification: http://bit.ly/DDI-CDI-webinar-series
- **Workshop on DDI-CDI, Friday 24 March 2023 https://bit.ly/DDI-CDI-Workshop**

WorldFAIR Case Studies

- **Chemistry** – making IUPAC assets FAIR
- **Nanomaterials** – applying NanoInchi and FAIR recommendations in Nanosafety.
- **Geochemistry** – recommendations for FAIR in geochemistry, particularly vocabularies.
- **Social Surveys Data** – data harmonisation between ESS and AussiESS.
- **Population Health –** INSPIRE - Integration of population surveys with clinical and genomics data for COVID-19 research in eastern and southern Africa.
- **Urban Health** – terminologies and making urban health data FAIR
- **Biodiversity** – improving GBIF data model in collaboration with TDWG - GBIF (Global Biodiversity Information Facility)
- **Agricultural Biodiversity** – pollinator data (KALRO, Embrapa, Meise, HiveTracks)
- **Ocean Science** – Implementing FAIR in the ODIS (Ocean Data and Information System) for the UNESCO Oceans' decade.
- **Disaster Risk Reduction** – recommendations on making DRR data and terminologies FAIR, case studies in Africa and Pacific Islands
- **Cultural Heritage** – recommendations on making cultural heritage data FAIR (particularly digital representation of heritage artefacts)



WorldFAIR Coordination and Synthesis

- **CODATA leading coordination and synthesis WP.**
- **D2.1: FAIR Implementation Profiles** for each case study (a methodology for understanding FAIR practices).
  - Workshop as part of the FAIR Week events in Leiden: https://bit.ly/WorldFAIR-FIPs-Workshop
  - Report 'FAIR Implementation Profiles (FIPs) in WorldFAIR: What Have We Learnt?': https://doi.org/10.5281/zenodo.7378109
- **D2.2:** Recommendations and documentation for a **Cross-Domain Interoperability Framework**. Working Group and Advisory Group have started contribution.
  - **Working Documents: https://doi.org/10.5281/zenodo.7652742**
  - **Workshop on 20 March https://bit.ly/CDIF-workshop**
  - **WorldFAIR Policy Brief: https://doi.org/10.5281/zenodo.7853170**
- **D2.3:** Findings will lead into recommendations for **domain sensitive FAIR assessment**.

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Thank you for your attention

Simon Hodson, CODATA
www.codata.org
simon@codata.org
@simonhodson99 ; @CODATAnews

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